

Request for Proposal (RFP)

Vector Database Ingestion Performance & Validation Test

Issue Date: _____

Proposal Due Date: _____

Prepared By: _____

1. Purpose

The purpose of this Request for Proposal (RFP) is to evaluate vector database platforms based on their ingestion capabilities, scalability, reliability, performance, operational simplicity, and production readiness.

Selected vendors will participate in a controlled ingestion benchmark using identical datasets and evaluation criteria.

2. Objectives

- Evaluate ingestion throughput.
- Measure indexing performance.
- Validate metadata ingestion.
- Measure search readiness after ingestion.
- Evaluate operational monitoring capabilities.
- Assess recovery after interrupted ingestion.
- Measure ingestion cost and resource utilization.

3. Test Dataset

Dataset	Description
Vector Count	10 Million vectors
Dimensions	768 Float32
Metadata	10-20 metadata fields per vector
Embedding Source	OpenAI / BAAI / Sentence Transformers
File Format	Parquet, JSONL, CSV

4. Scope of Testing

4.1 Bulk Ingestion

- Initial data load
- Batch ingestion
- Streaming ingestion
- Parallel ingestion

4.2 Incremental Updates

- Insert
- Update
- Delete
- Upsert

4.3 Index Creation

- Automatic indexing
- Background indexing
- Manual indexing

5. Technical Requirements

Requirement	Expectation
API	REST, gRPC, or SDK
Supported Languages	Python preferred
Metadata Filtering	Required
Horizontal Scaling	Required
Cloud Deployment	AWS / Azure / GCP
Container Support	Docker / Kubernetes
Authentication	OAuth/API Keys/RBAC

6. Test Scenarios

Scenario	Description
S1	Cold bulk ingestion
S2	Parallel ingestion using multiple workers
S3	Continuous streaming ingestion
S4	Recovery after node failure
S5	Metadata-heavy ingestion
S6	Mixed insert/update/delete workload

7. Performance Metrics

Metric	Measurement
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Vectors/sec	Average ingestion throughput
Total Load Time	Time until searchable
CPU Usage	Average utilization
Memory Usage	Peak RAM
Disk Usage	Total storage consumed
Index Build Time	Completion duration
Recovery Time	Restart after interruption
Error Rate	Failed ingestion percentage

8. Deliverables

Vendors shall provide:

- Deployment architecture
- Installation guide
- Ingestion scripts
- Performance report
- Resource utilization report
- Operational monitoring dashboard
- Documentation
- Support contacts

9. Evaluation Criteria

Category	Weight
Ingestion Performance	30%

Scalability	20%
Operational Simplicity	15%
Reliability	15%
Monitoring & Observability	10%
Documentation & Support	10%

10. Vendor Response Template

Question	Vendor Response
Maximum supported vectors?	
Recommended hardware?	
Average ingestion throughput?	
Index type used?	
Supports streaming?	
Supports updates?	
Supports deletes?	
High availability?	
Disaster recovery?	
Monitoring tools?	
Licensing model?	
Estimated infrastructure cost?	

11. Acceptance Criteria

- 100% successful ingestion of all records.
- No data corruption.
- Search available immediately after indexing.
- Metadata accuracy verified.
- System recovery demonstrated after interruption.

- Performance meets agreed SLA.

12. Timeline

Milestone	Date
RFP Released	_____
Questions Due	_____
Proposal Submission	_____
Vendor Demonstrations	_____
Performance Testing	_____
Selection	_____

13. Contact Information

Organization:

Contact Name:

Email:

Phone:

Address:

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